

Polynomial Graphs: End Behavior and Turning Points

A reasoning worksheet for analyzing the general shape of polynomial functions

Big Idea

For a polynomial function, the **leading term** tells us the long-term direction of the graph.

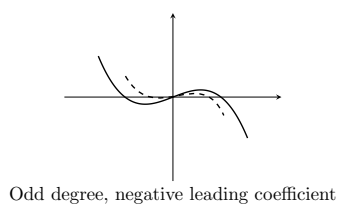
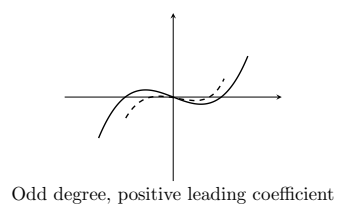
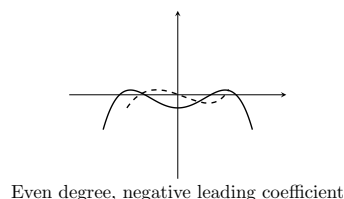
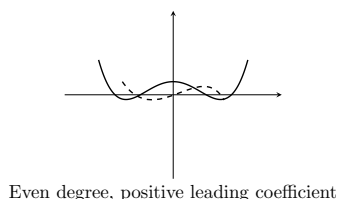
$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

The key features are:

- the degree, n
- whether n is even or odd
- whether the leading coefficient, a_n , is positive or negative

Lower-degree terms may change the middle of the graph, but they do not control the end behavior.

The Four Main End-Behavior Cases



End Behavior Summary

Degree	Leading Coefficient	As $x \rightarrow -\infty$	As $x \rightarrow \infty$
Even	Positive	$f(x) \rightarrow \infty$	$f(x) \rightarrow \infty$
Even	Negative	$f(x) \rightarrow -\infty$	$f(x) \rightarrow -\infty$
Odd	Positive	$f(x) \rightarrow -\infty$	$f(x) \rightarrow \infty$
Odd	Negative	$f(x) \rightarrow \infty$	$f(x) \rightarrow -\infty$

Important Fact

A polynomial of degree n can have at most

$$n - 1$$

turning points.

This does **not** mean it must have $n - 1$ turning points. It means that is the maximum possible.

Part A: Warm-Up

For each polynomial, identify the degree, the sign of the leading coefficient, and the end behavior.

1. $f(x) = 3x^5 - 2x^2 + 1$

Degree: _____ Leading coefficient sign: _____

Left end: _____ Right end: _____

2. $g(x) = -4x^8 + x^3 - 7$

Degree: _____ Leading coefficient sign: _____

Left end: _____ Right end: _____

3. $h(x) = -x^7 + 5x^6 - 3$

Degree: _____ Leading coefficient sign: _____

Left end: _____ Right end: _____

4. $p(x) = 9x^4 - 12x + 1$

Degree: _____ Leading coefficient sign: _____

Left end: _____ Right end: _____

Part B: Match the Polynomial

Match each polynomial to one of the four end-behavior types.

1. $f(x) = x^9 - 2x^4 + 3$

2. $g(x) = -3x^{10} + 7x$

3. $h(x) = 5x^6 - 2x^5 + 1$

4. $p(x) = -2x^3 + 9$

Type	Description
A	Both ends up
B	Both ends down
C	Left down, right up
D	Left up, right down

Part C: Reverse Thinking

A polynomial has both ends rising and exactly three turning points.

- What can you determine about the degree?
- What are the possible degrees?
- Explain your reasoning.

Part D: Possible or Impossible?

For each statement, decide whether it is possible. If possible, give an example. If impossible, explain why.

1. A polynomial has odd degree and both ends up.
2. A polynomial has even degree and exactly one turning point.
3. A polynomial has degree 3 and four turning points.
4. A polynomial has negative leading coefficient and left end up.

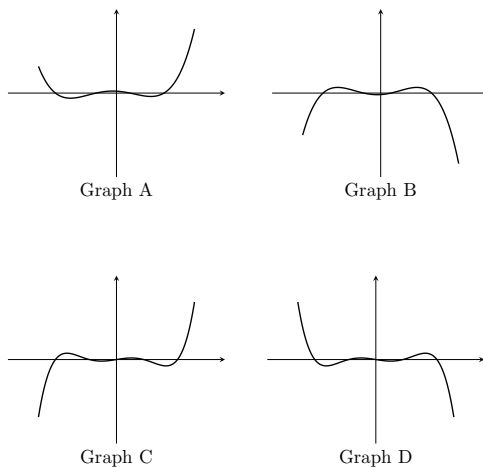
Part E: Mystery Polynomial

A polynomial has opposite end behaviors and at most four turning points.

- Must the degree be even or odd?
- What are the possible degrees?
- Explain why.

Part F: Graph Detective

Study the four graphs below.



Answer the questions.

- Which graph or graphs could have even degree?
- Which graph or graphs could have odd degree?
- Which graph or graphs have positive leading coefficient?
- Which graph or graphs have negative leading coefficient?
- Which graph could represent a polynomial of degree 5?

Part G: Find the Error

A student says:

“This graph has three turning points, so the polynomial must be degree 4.”

Is the student correct? Explain carefully.

Part H: Create Your Own

Construct examples of polynomial functions with the requested features.

1. An even-degree polynomial with positive leading coefficient and exactly two turning points.
2. An odd-degree polynomial with negative leading coefficient and exactly four turning points.

Part I: Reasoning Challenge

Can a polynomial have both ends up and cross the x -axis exactly once?

Explain. If possible, give an example.

Part J: The Big Synthesis

Four polynomials are hidden. For each one, sketch a possible graph and write one possible equation.

Polynomial	Degree Parity	Leading Coefficient Sign	Maximum Turning Points
A	Odd	Positive	4
B	Even	Negative	5
C	Odd	Negative	6
D	Even	Positive	7

1. Sketch a possible graph for Polynomial A. Write a possible equation.

2. Sketch a possible graph for Polynomial B. Write a possible equation.

3. Sketch a possible graph for Polynomial C. Write a possible equation.

4. Sketch a possible graph for Polynomial D. Write a possible equation.

Extension: Minimum Possible Degree

For each graph description, determine the minimum possible degree.

Description	Minimum Possible Degree	Reason
Both ends up, 2 turning points		
Both ends down, 5 turning points		
Left down, right up, 4 turning points		
Left up, right down, 3 turning points		
Both ends up, 0 turning points		
Opposite end behavior, 6 turning points		

Final Reflection

A polynomial has degree 20.

- What can you determine with certainty about its graph?
- What can you not determine?